

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\xi_k^\alpha(x) &= \left(\left(\frac{1}{R(A(x))R(A(x))} \right)^{1/4} \right)^{\alpha\alpha'} B_{kk'}^{\alpha'}(x) \\ &= \frac{1}{\pi\sqrt{2}} \int_0^\infty d\lambda \lambda^{-1/4} \frac{1}{\lambda + R \bullet R} \bullet B(x)\end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}s &= \sum_{i=0}^{r-1} \left\{ \left\langle \frac{-2np^i}{q-1} \right\rangle + \left\langle \frac{ndp^i}{q-1} \right\rangle + \left\langle \frac{-n(d-1)p^i}{q-1} \right\rangle - \left\langle \frac{-np^i}{q-1} \right\rangle \right\} \\ &= \sum_{i=0}^{r-1} \left\{ \left\lfloor \frac{-np^i}{q-1} \right\rfloor - \left\lfloor \frac{-2np^i}{q-1} \right\rfloor - \left\lfloor \frac{ndp^i}{q-1} \right\rfloor - \left\lfloor \frac{-n(d-1)p^i}{q-1} \right\rfloor \right\},\end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\frac{\partial \eta_k(\xi)}{\partial \xi} &= \frac{\sigma_k}{\xi^2} + \frac{2 \log(C_k)}{\xi^3} + \frac{k}{\xi^3} + \frac{k}{\xi^2 \sigma_k} \\ &\quad - \left(\frac{1}{2\sigma_k} + \frac{2k \log(\sigma_k + \xi)}{\xi^3} + \frac{\log\left(\frac{1}{2} \left(1 + \frac{\xi}{\sigma_k}\right)\right)}{\xi^3} + \frac{\xi^2 - \sigma_k^2}{2\xi^2 \sigma_k^2 (\sigma_k + \xi)} \right).\end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}S_{\omega, \mathbf{c}}(U) &:= E(U) + \omega Q(U) + \mathbf{c} \cdot \mathbf{P}(U), \\ K_{\omega, \mathbf{c}}(U) &:= \partial_\lambda S_{\omega, \mathbf{c}}(\lambda U)|_{\lambda=1} = 2L(U) + 3N(U) + 2\omega Q(U) + 2\mathbf{c} \cdot \mathbf{P}(U), \\ L_{\omega, \mathbf{c}}(U) &:= K_{\omega, \mathbf{c}}(U) - 3N(U) = 2L(U) + 2\omega Q(U) + 2\mathbf{c} \cdot \mathbf{P}(U).\end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$U'_{[4] \times [N]} = - \begin{pmatrix} U'_{11} & U'_{12} & \cdots & U'_{1N} \\ U'_{21} & U'_{22} & \cdots & U'_{2N} \\ U'_{31} & U'_{32} & \cdots & U'_{3N} \\ U'_{41} & U'_{42} & \cdots & U'_{4N} \end{pmatrix},$$